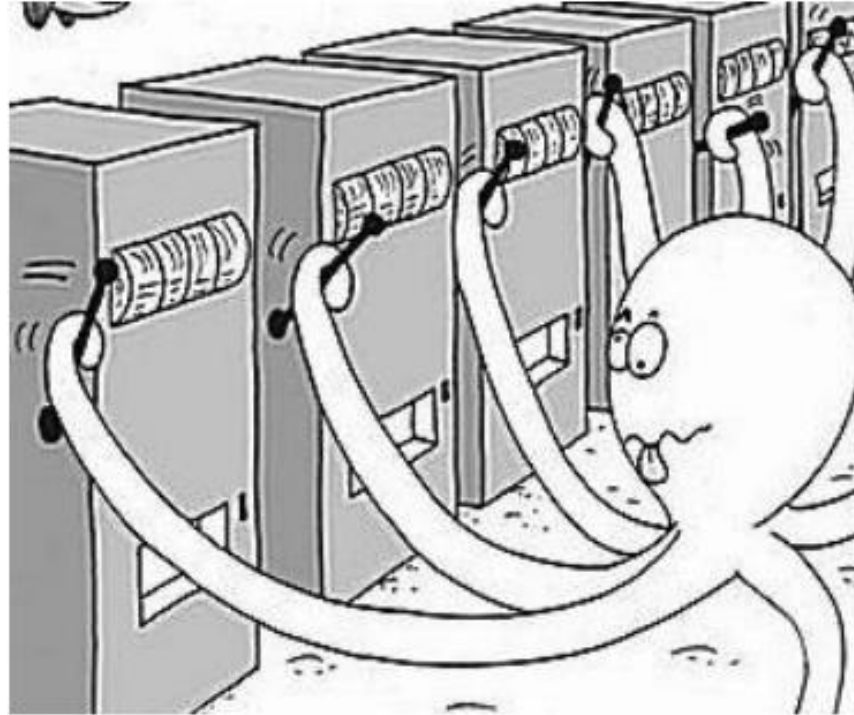


16-831: Introduction to Robot Learning, Fall 2024

Bandits and Preference-based Learning



Instructor: Guanya Shi

Recording Reminder

- ☐ This session will be audio recorded for educational use by other students in this course.

Course Structure

Latest schedule: <https://16-831.github.io/fall24/lectures>

- ☐ Topic group 1: Course introduction: What is robot learning and why?
- ☐ Topic group 2: Machine/Deep learning refresher
- ☐ Topic group 3: Imitation learning
- ☐ Topic group 4: Model-free reinforcement learning
- ☐ Topic group 5: Model-based reinforcement learning
- ☒ Topic group 6: Bandit, preference-based learning, exploration
- ☐ Topic group 7: Reinforcement learning from offline data (one guest lecture)
- ☐ Topic group 8: Specialized topics (one guest lecture)
- ☐ Project presentation

Outline

- ❑ Multi-armed bandits (MAB) problem
- ❑ Algorithms for MAB:
 - Epsilon-greedy
 - Upper confidence bound (UCB)
 - Thompson sampling
- ❑ Extension of MAB: contextual bandits
- ❑ Preference-based bandits: motivations

(Some slides are from 22F, 23F, Berkeley CS 285, CMU 10-703, and Yisong Yue)

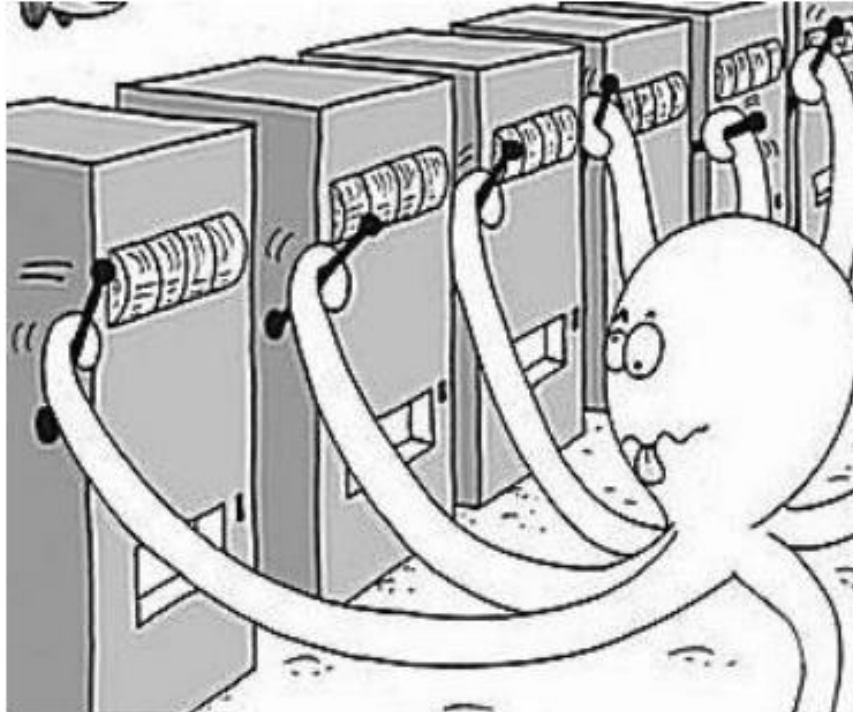
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Multi-Arm Bandits (MAB) Problem

- ❑ The simplest RL problem!
 - Still many open problems!
- ❑ NO state or transition/dynamics, just action!
 - K actions (a.k.a. arms or bandits)
 - Each action has an unknown average reward
 - Goal: find the “optimal” action with the largest average reward



Why Study Bandits Problem?

- ❑ Many real-world problems can be formulated as a bandits problem
 - Recommendation system
 - Experimental design
 - Robot learning
 - RLHF for LLM
- ❑ Allow us to understand the fundamental ideas behind RL:
 - The trade-off between **exploration** and **exploitation**
 - **Optimistic** and **pessimistic** mechanism
- ❑ Allow us to study variants of reward-based RL:
 - Preference-based feedback
- ❑ Allow us to get into RL theory:
 - Regret guarantee, etc

Multi-Arm Bandits (MAB) Problem: Formulations

- ❑ k actions (a.k.a. arms or bandits)
- ❑ Each action has an average reward: $\mu_1, \mu_2, \dots, \mu_k$
 - Unknown to us
 - Assume WLOG that μ_1 is largest
- ❑ For $t = 1, \dots, T$
 - Algorithm chooses action $a(t) \in [1, k]$
 - Receives random reward $R(t)$
 - Expectation of $R(t)$ is $\mu_{a(t)}$

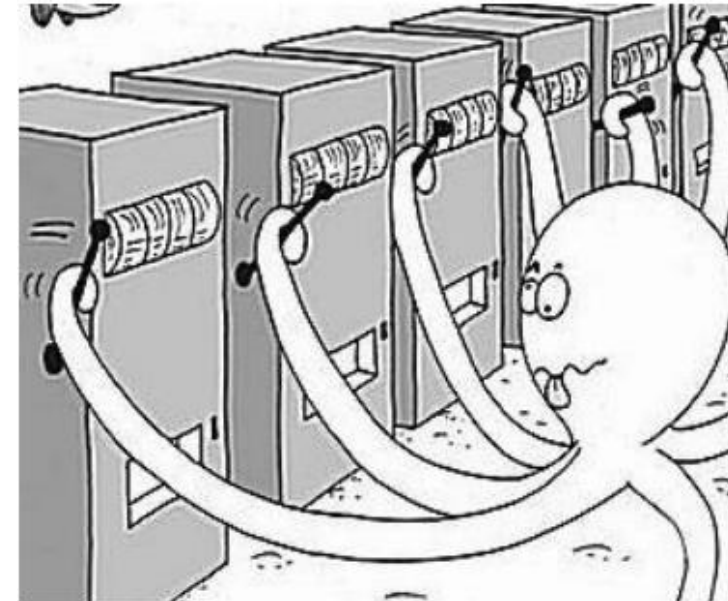
Algorithm only receives feedback on chosen action

- ❑ **Goal:** minimize **Regret** = $T\mu_1 - (\mu_{a(1)} + \mu_{a(2)} + \dots + \mu_{a(T)})$

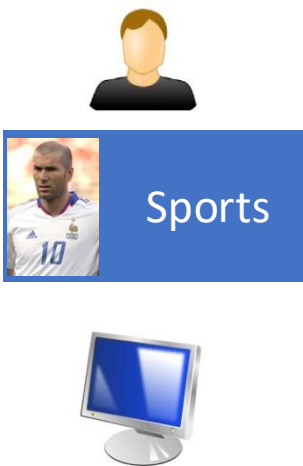
If we had perfect information to start

Expected reward of algorithm

- ❑ Why this problem is hard?
 - Counterfactual reasoning (we only have feedback from chosen action)
 - Randomness (we have to balance between exploration and exploitation)



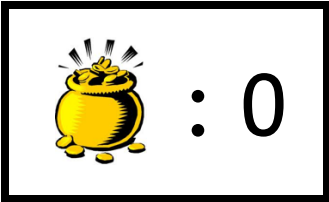
MAB Example: Interactive Personalization



Average Likes

Shown

				
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0	0	0	1	0

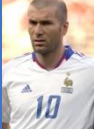


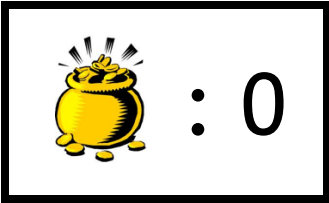
MAB Example: Interactive Personalization



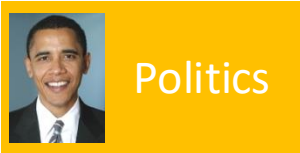
Average Likes

Shown

				
--	--	--	0	--
0	0	0	1	0



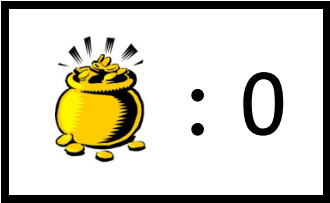
MAB Example: Interactive Personalization



Average Likes

Shown

				
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0	0	1	1	0



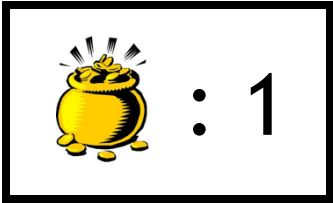
MAB Example: Interactive Personalization



Average Likes

Shown

					
Average Likes	--	--	1	0	--
# Shown	0	0	1	1	0



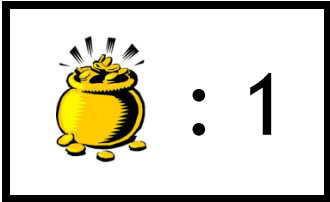
MAB Example: Interactive Personalization



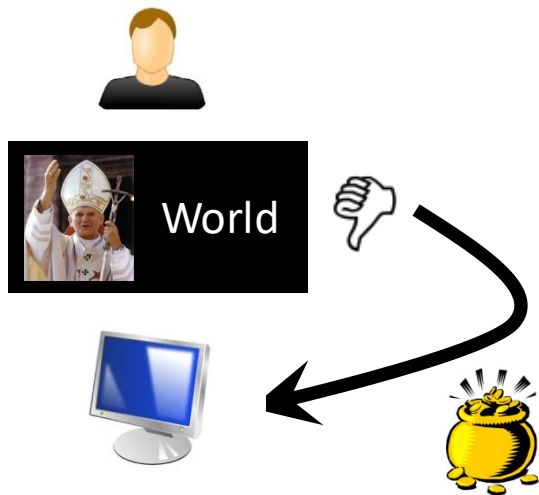
Average Likes

Shown

--	--	1	0	--
0	0	1	1	1



MAB Example: Interactive Personalization



Average Likes

Shown

				
--	--	1	0	0
0	0	1	1	1



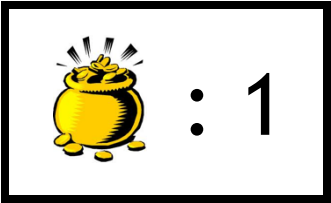
MAB Example: Interactive Personalization



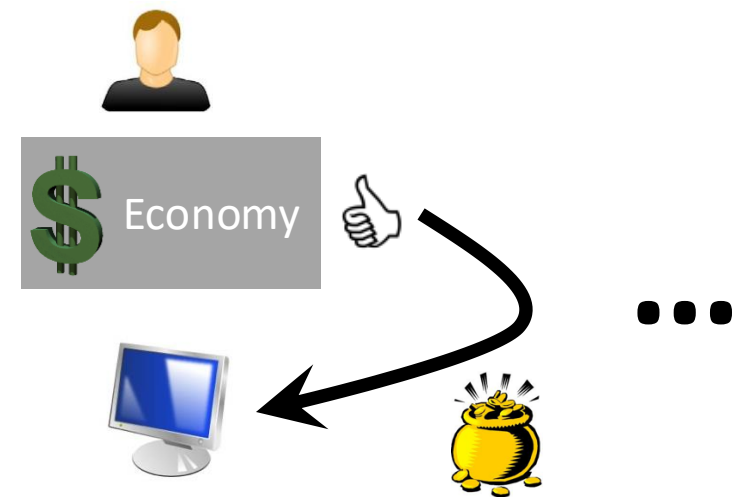
Average Likes

Shown

				
--	--	1	0	0
0	1	1	1	1

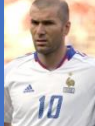


MAB Example: Interactive Personalization



Average Likes

Shown

				
--	1	1	0	0
0	1	1	1	1

 : 2

MAB Example: Interactive Personalization

❑ What should algorithm recommend?

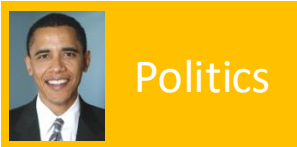
Exploit:



Explore:



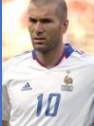
Best:

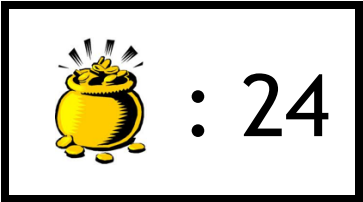


How to Optimally Balance Explore/Exploit Tradeoff?
Characterized by the Multi-Armed Bandit Problem

Average Likes

Shown

					
Average Likes	--	0.44	0.4	0.33	0.2
# Shown	0	25	10	15	20



The Goal of MAB: Minimize Regret

- OPT (optimal): what we would have done if we had perfect information in the beginning
- ALG (algorithm): what the algorithm chose to do

$$\text{💰}(OPT) = \text{💰}(\text{Obama}) + \text{💰}(\text{Obama}) + \text{💰}(\text{Obama}) \dots$$

$$\text{💰}(ALG) = \text{💰}(\text{Messi}) + \text{💰}(\text{Obama}) + \text{💰}(\text{Pope}) \dots$$

Time Horizon

Regret: $R(T) = \text{💰}(OPT) - \text{💰}(ALG)$

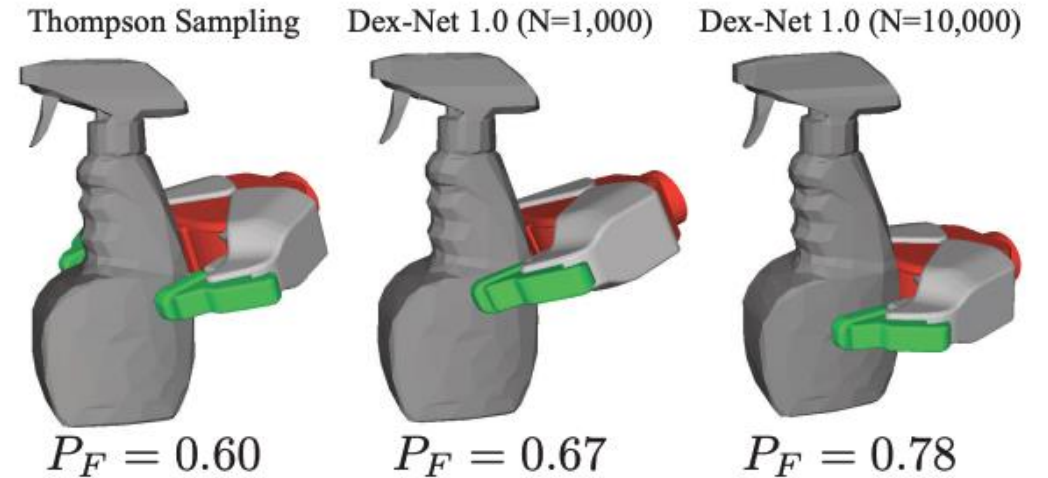
- I.e., opportunity cost of not knowing preferences
- “No-regret” (or sublinear regret) if $\frac{R(T)}{T} \rightarrow 0$. E.g., $R(T) = \log T$, $R(T) = \sqrt{T}$
 - I.e., asymptotically, the “mistakes” are negligible!

MABs in Robot Learning

❑ ICRA 2016

Dex-Net 1.0: A Cloud-Based Network of 3D Objects for Robust Grasp Planning Using a Multi-Armed Bandit Model with Correlated Rewards

Jeffrey Mahler¹, Florian T. Pokorny¹, Brian Hou¹, Melrose Roderick¹, Michael Laskey¹, Mathieu Aubry¹,
Kai Kohlhoff², Torsten Kröger², James Kuffner², Ken Goldberg¹



❑ ICRA 2020 (best paper award)

Preference-Based Learning for Exoskeleton Gait Optimization

Maegan Tucker^{*1}, Ellen Novoseller^{*2}, Claudia Kann¹,
Yanan Sui³, Yisong Yue², Joel W. Burdick^{1,2}, and Aaron D. Ames^{1,2}



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(Some slides are from 22F, 23F, Berkeley CS 285, CMU 10-703, and Yisong Yue)

A Simple Algorithm: epsilon-Greedy

- ❑ $Q(a)$: value/reward estimation of action a
- ❑ $N(a)$: number of times the action a has been selected
- ❑ $Q(a)$ is just a numerical mean:

$$Q_t(a) \doteq \frac{\text{sum of rewards when } a \text{ taken prior to } t}{\text{number of times } a \text{ taken prior to } t} = \frac{\sum_{i=1}^{t-1} R_i \cdot \mathbf{1}_{A_i=a}}{\sum_{i=1}^{t-1} \mathbf{1}_{A_i=a}}$$

A simple bandit algorithm

Initialize, for $a = 1$ to k :

$Q(a) \leftarrow 0$ Action value estimates

$N(a) \leftarrow 0$ Number of times action has been selected

Repeat forever: Greedy action

$A \leftarrow \begin{cases} \arg \max_a Q(a) & \text{with probability } 1 - \varepsilon \\ \text{a random action} & \text{with probability } \varepsilon \end{cases}$ (breaking ties randomly)

$R \leftarrow \text{bandit}(A)$ Receive Reward

$N(A) \leftarrow N(A) + 1$ Update Counts

$Q(A) \leftarrow Q(A) + \frac{1}{N(A)} [R - Q(A)]$ Update action value estimates

exploitation

exploration

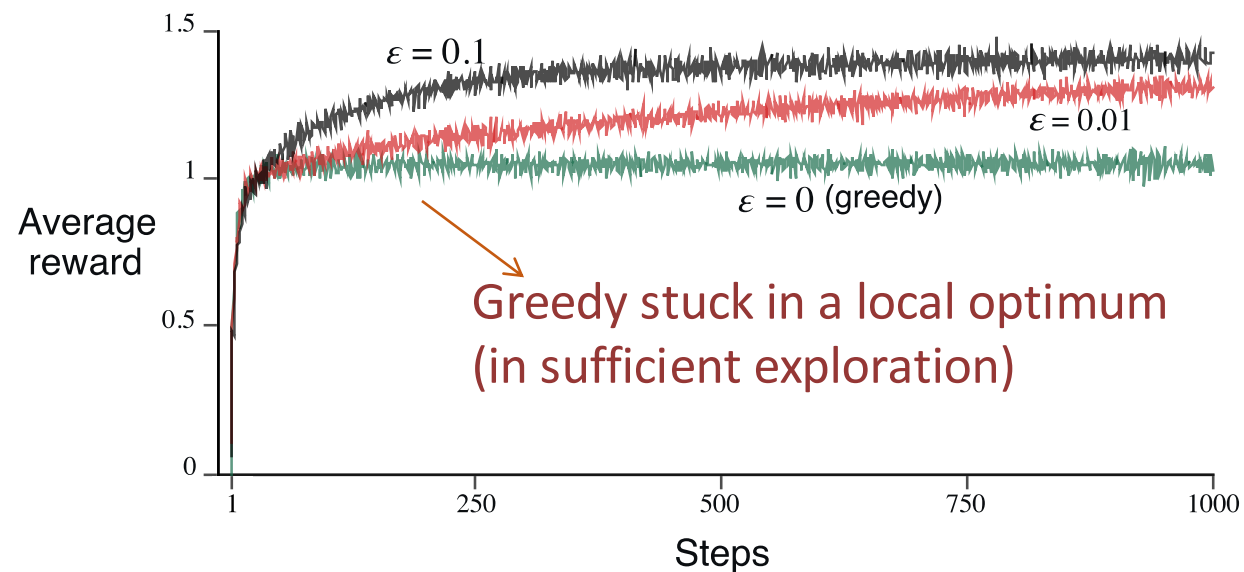
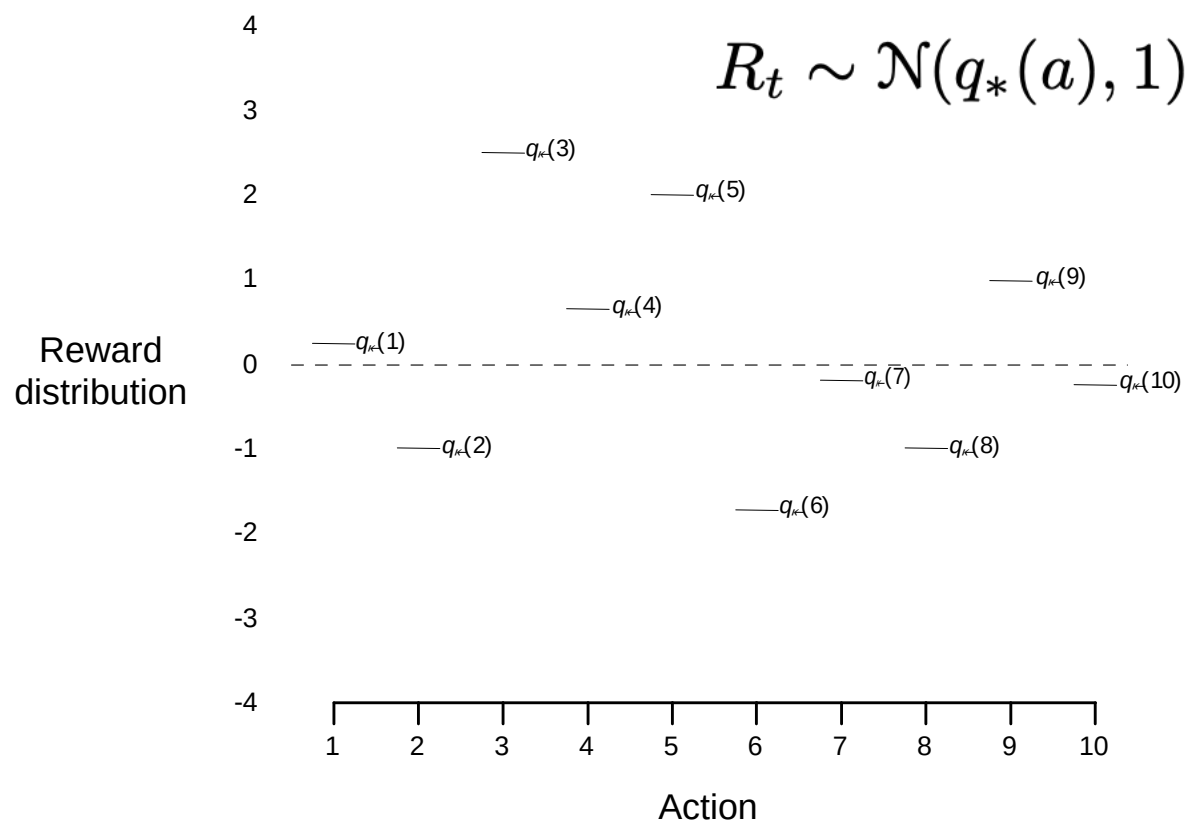
- ❑ You can do both exploitation and exploration, but NOT at the same time!

A Simple Algorithm: epsilon-Greedy

- ❑ 10-armed testbed
- ❑ Randomize the testbed 2000 times

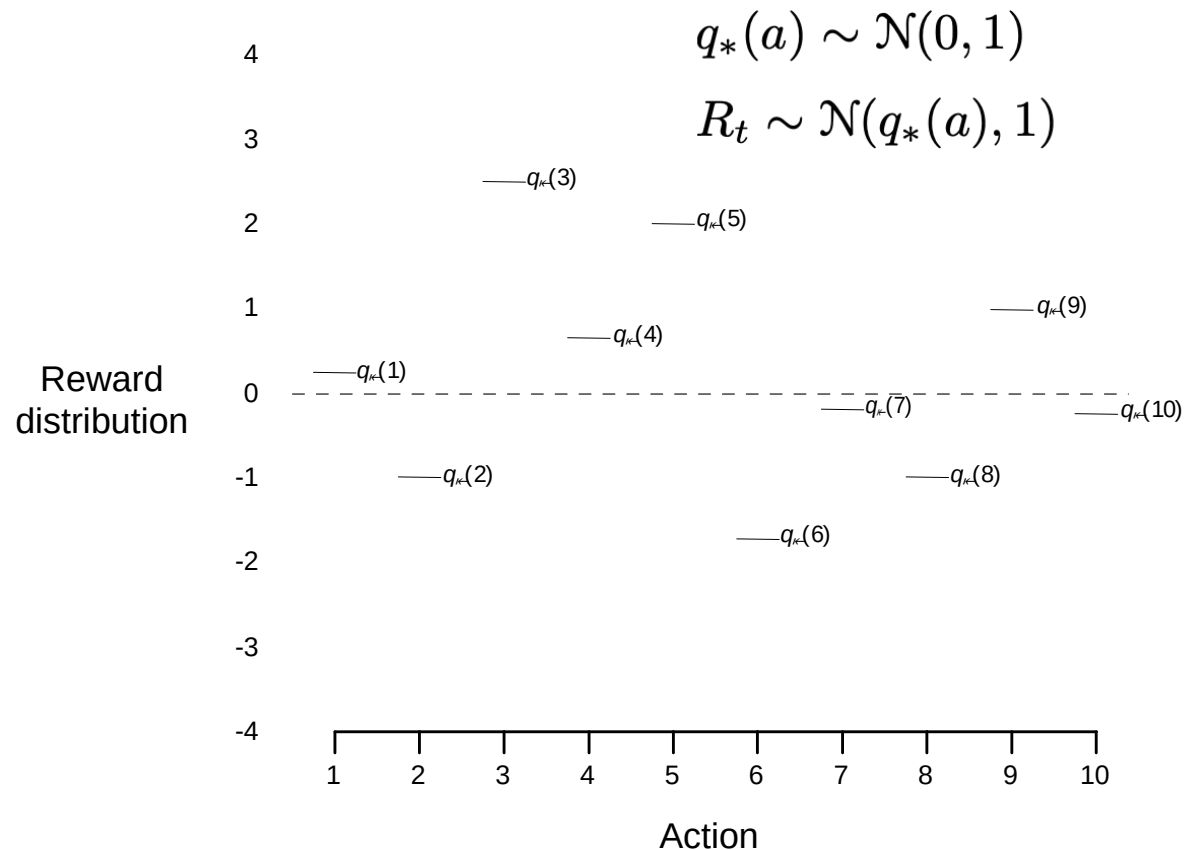
$$q_*(a) \sim \mathcal{N}(0, 1)$$

$$R_t \sim \mathcal{N}(q_*(a), 1)$$

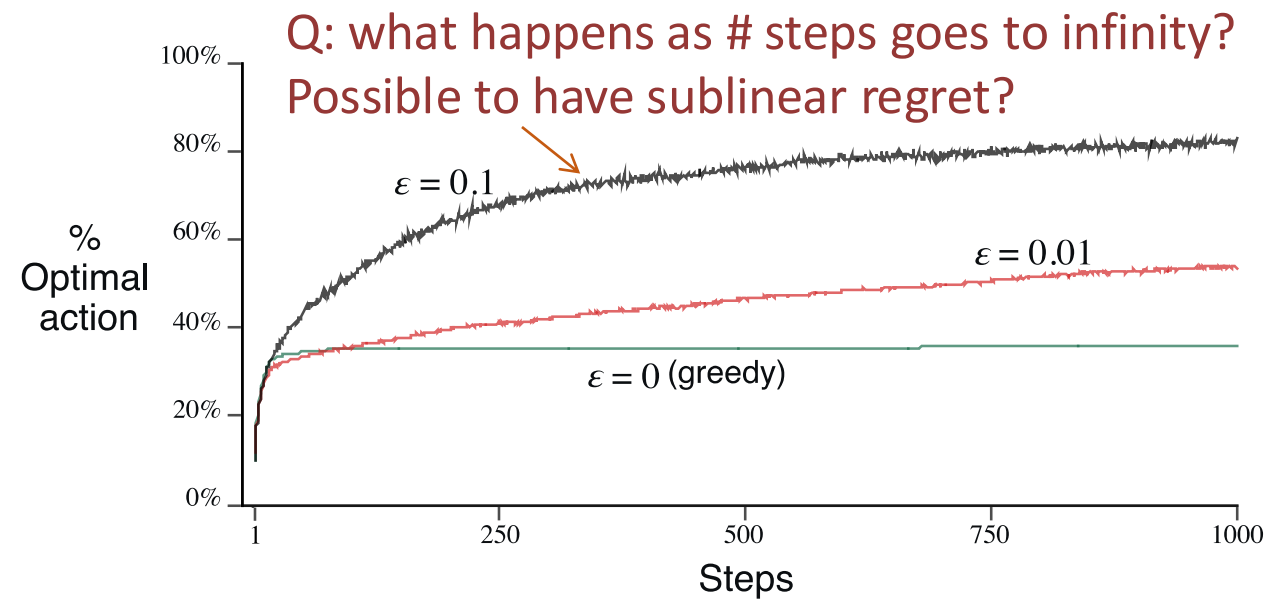


A Simple Algorithm: epsilon-Greedy

- ❑ 10-armed testbed
- ❑ Randomize the testbed 2000 times



% of tasks in which optimal action is selected



What's Wrong with epsilon-Greedy?

- ❑ The exploration part is too random: with probability epsilon, we explore an action **uniformly at random!**
- ❑ This is bad if some arms are much worse than others (not worth exploring them again!)
- ❑ How to fix it: more strategic exploration that adapts!

A simple bandit algorithm

Initialize, for $a = 1$ to k :

$Q(a) \leftarrow 0$ Action value estimates

$N(a) \leftarrow 0$ Number of times action has been selected

Repeat forever: Greedy action

$A \leftarrow \begin{cases} \arg \max_a Q(a) & \text{with probability } 1 - \varepsilon \\ \text{a random action} & \text{with probability } \varepsilon \end{cases}$ (breaking ties randomly)

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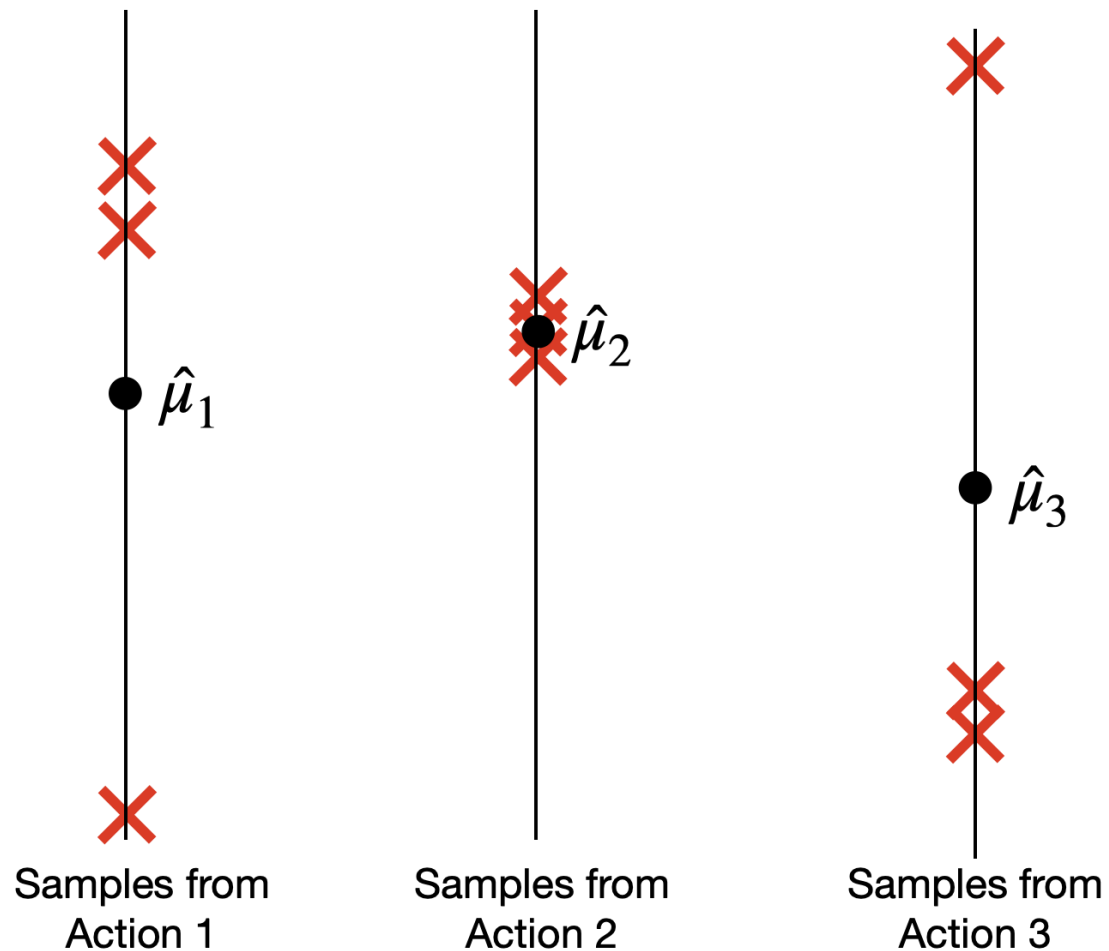
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exploitation

exploration

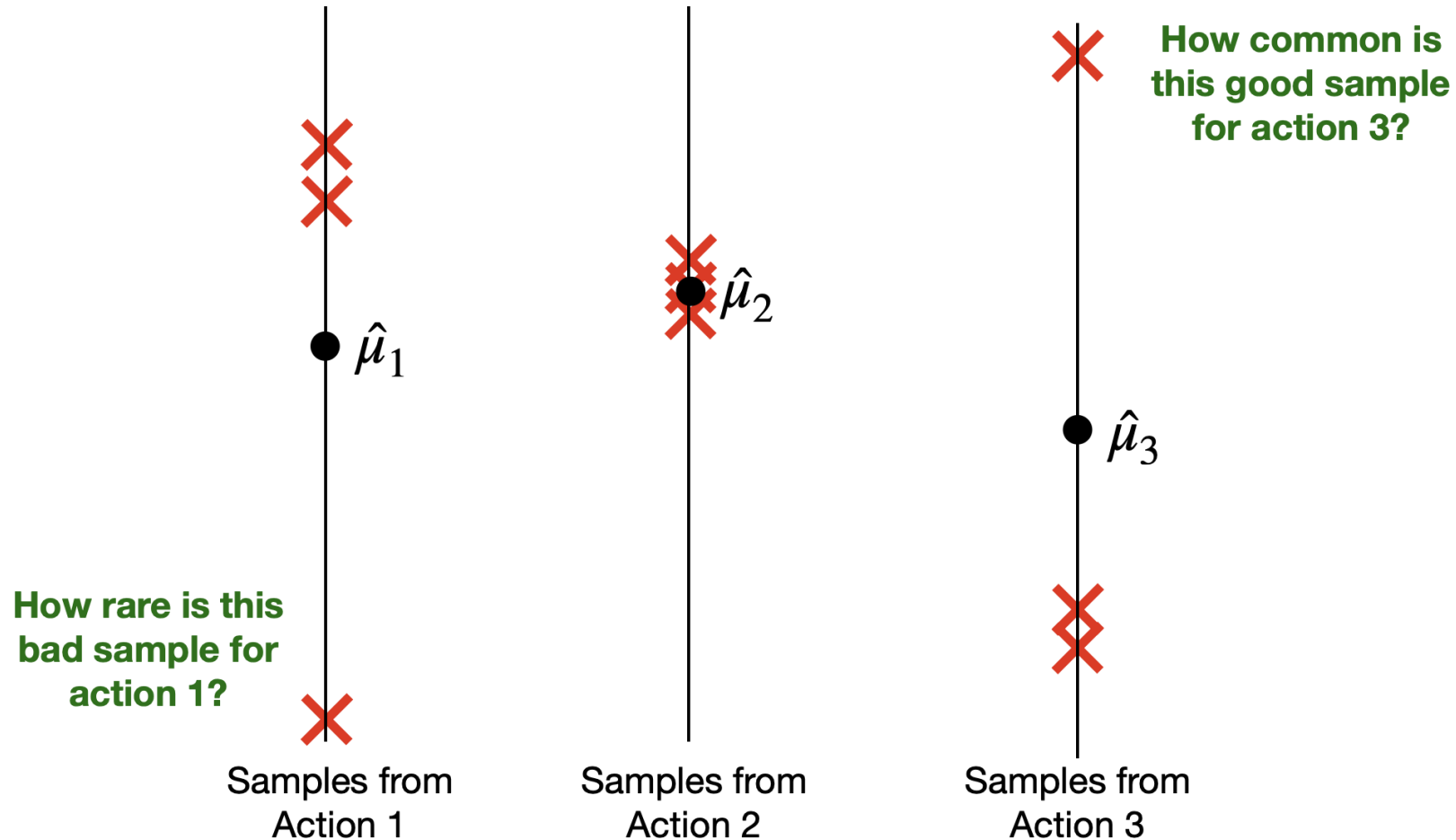
Motivating Examples

Which of these actions would you pick?



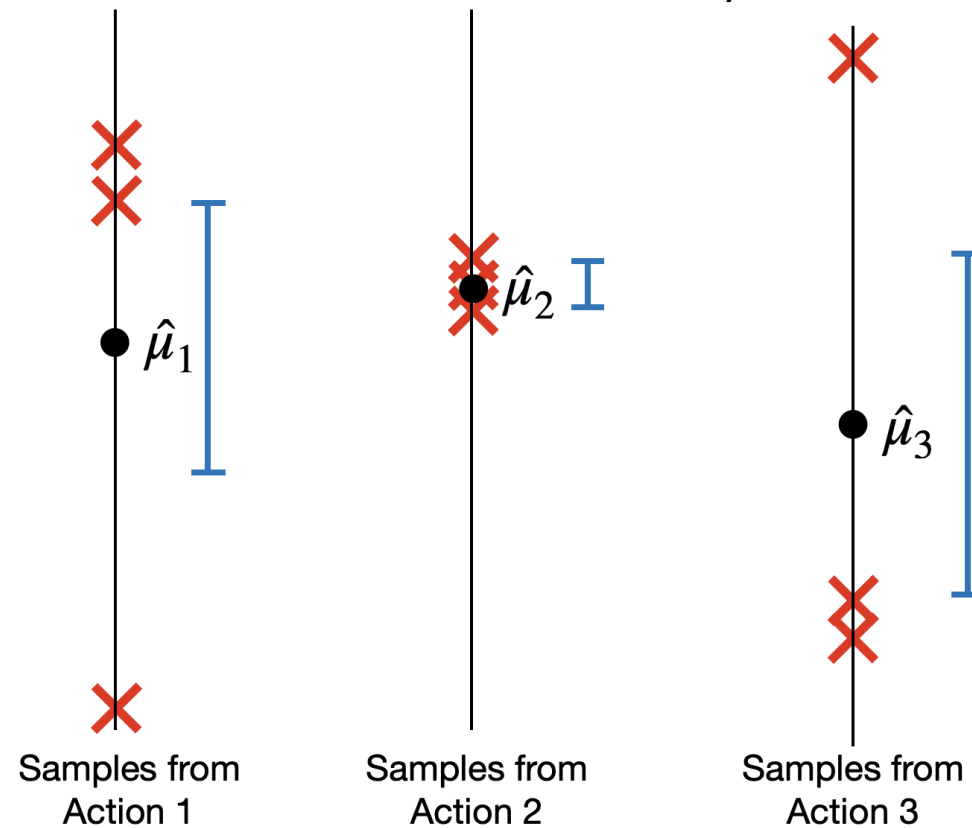
Motivating Examples

There is some additional information that we need to know...



Motivating Examples

Instead of estimating a single $\hat{\mu}$ it might be better to estimate a range of possible values for $\hat{\mu}$



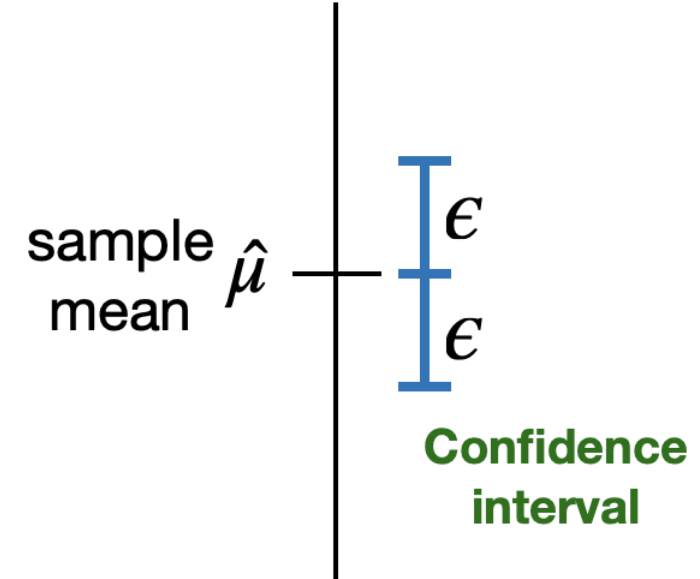
This range represents our uncertainty in not knowing the true value of the mean μ

Motivating Examples

- ❑ How to get the confidence interval?
 - Law of large numbers and concentration inequality!
- ❑ Let's assume that the reward distributions for each action are given by a Gaussian with a known variance σ and an unknown mean μ (different mean for each action)
- ❑ Hoeffding's inequality:

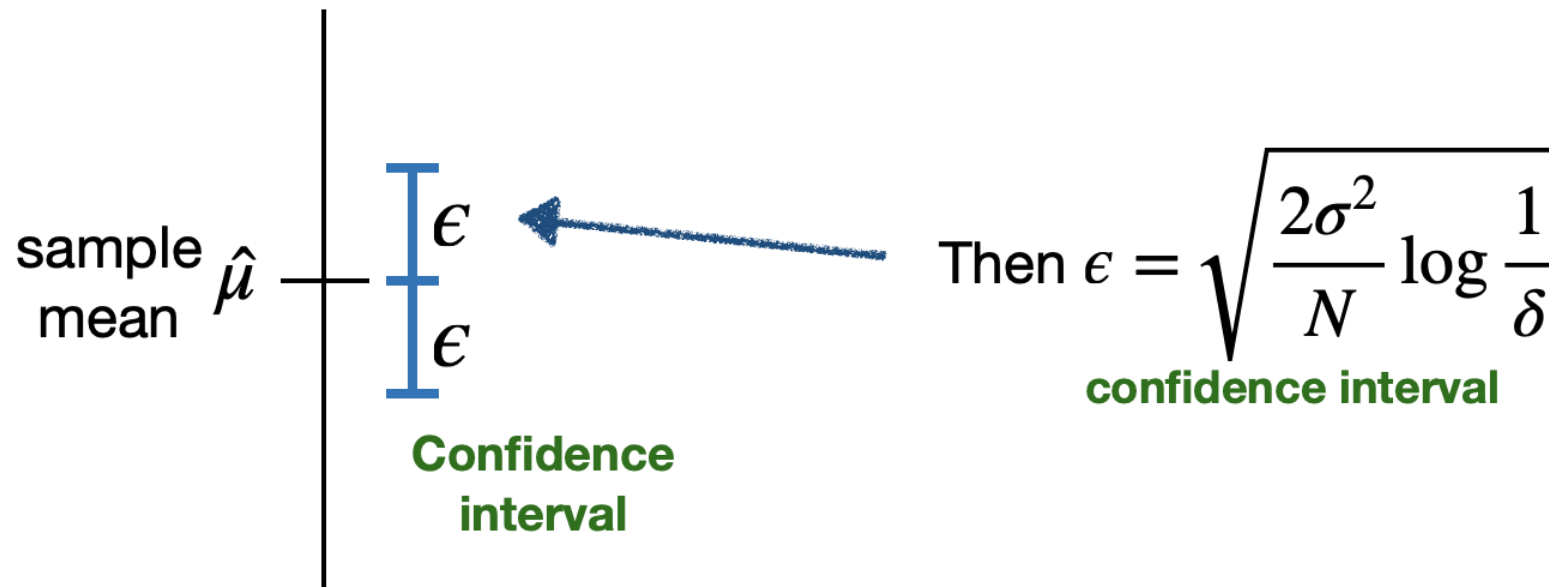
$$P(|\hat{\mu} - \mu| \leq \epsilon) \geq 1 - 2e^{-N\epsilon^2/(2\sigma^2)}$$

sample mean lies within ϵ of the true mean confidence



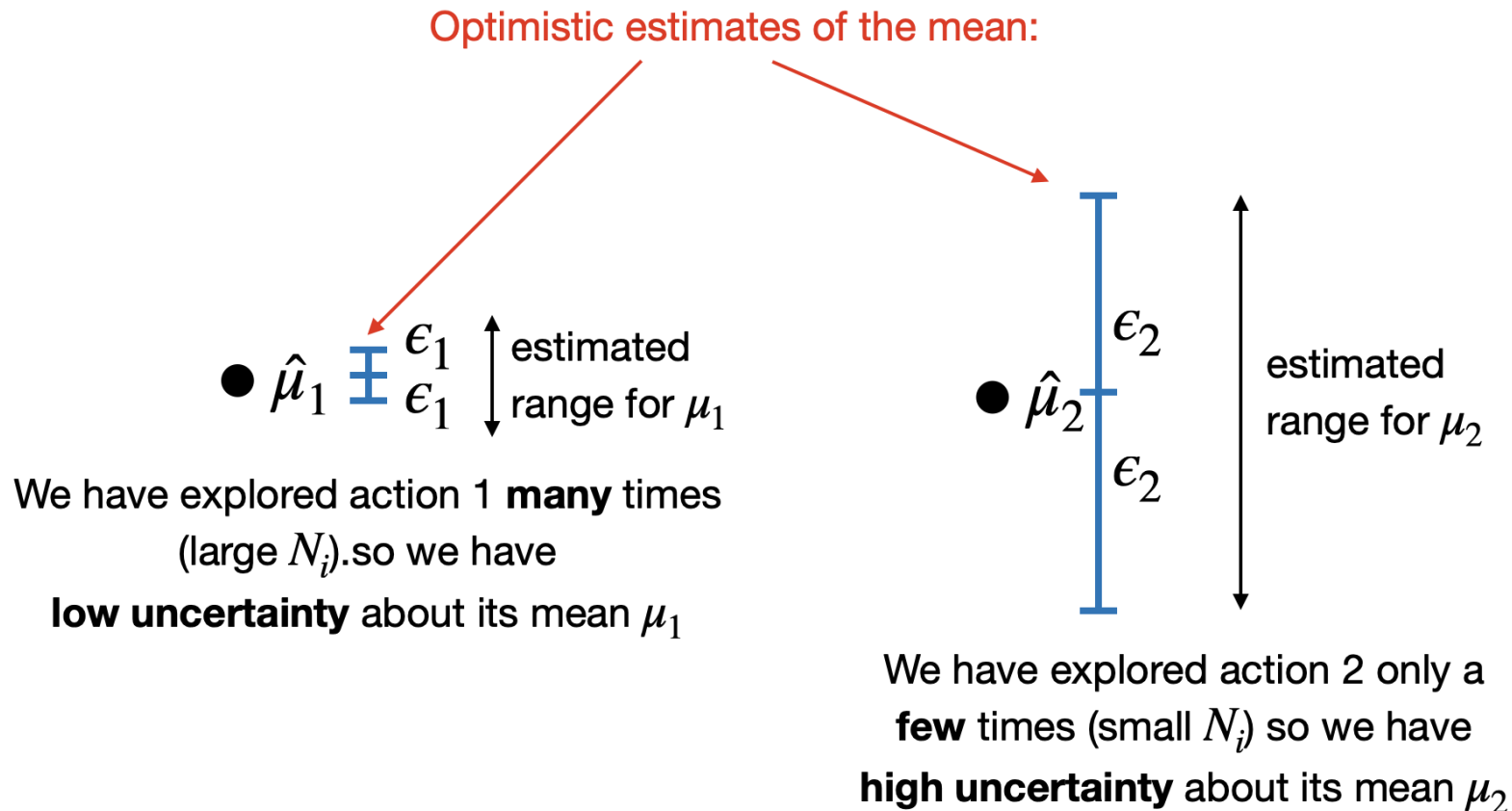
Motivating Examples

- ❑ How to get the confidence interval?
 - Law of large numbers and concentration inequality!
- ❑ Let's assume that the reward distributions for each action are given by a Gaussian with a known variance σ and an unknown mean μ (different mean for each action)
- ❑ Then we can have the following confidence interval with probability at least $1 - 2\delta$
 - Good news: decrease as N increases (i.e., concentration)!



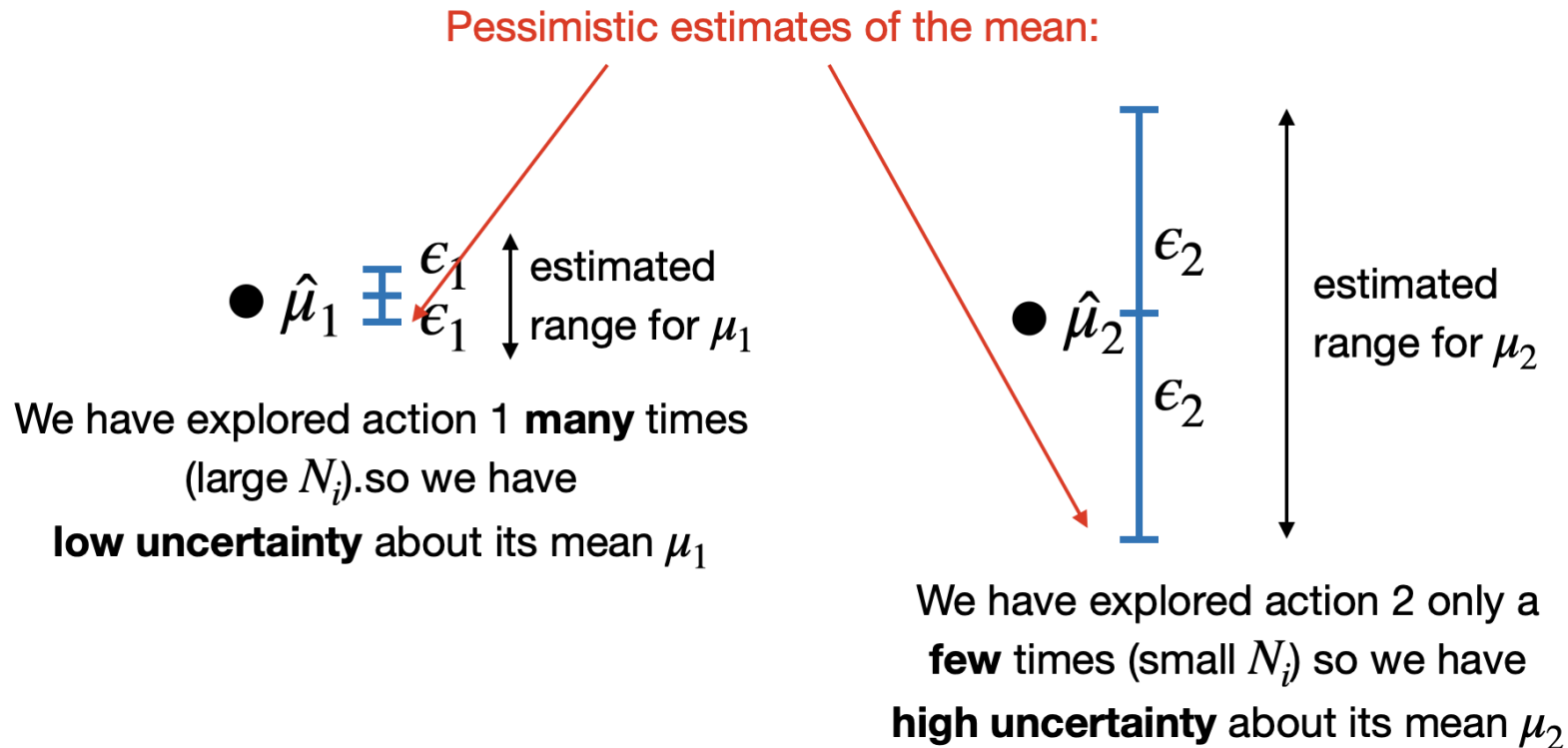
UCB: Upper Confidence Bounds

- ❑ Key idea: choose the most optimistic action, based on $\hat{\mu} + \epsilon$
- ❑ Principles: **optimism in the face of uncertainty** (OFU)



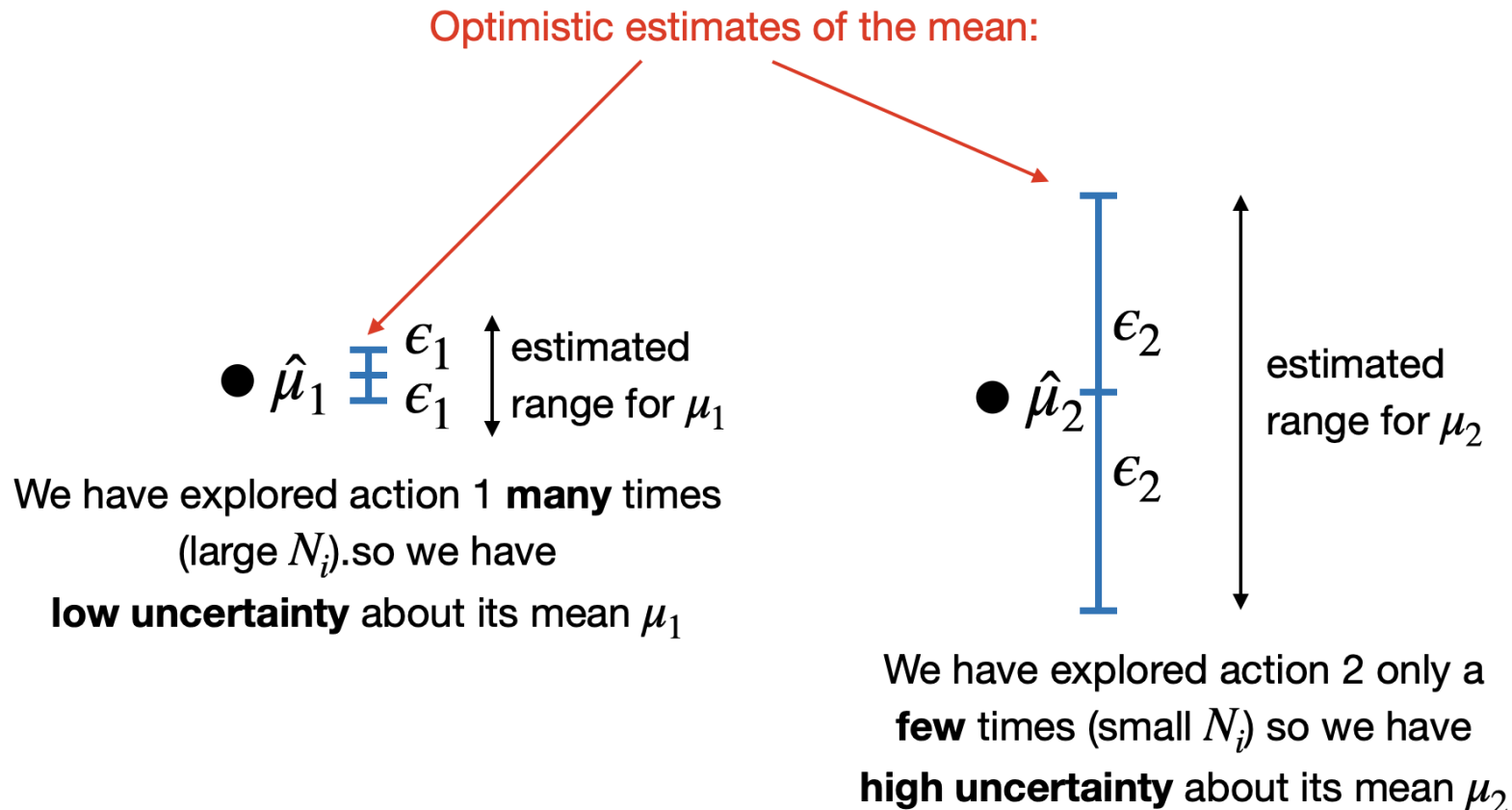
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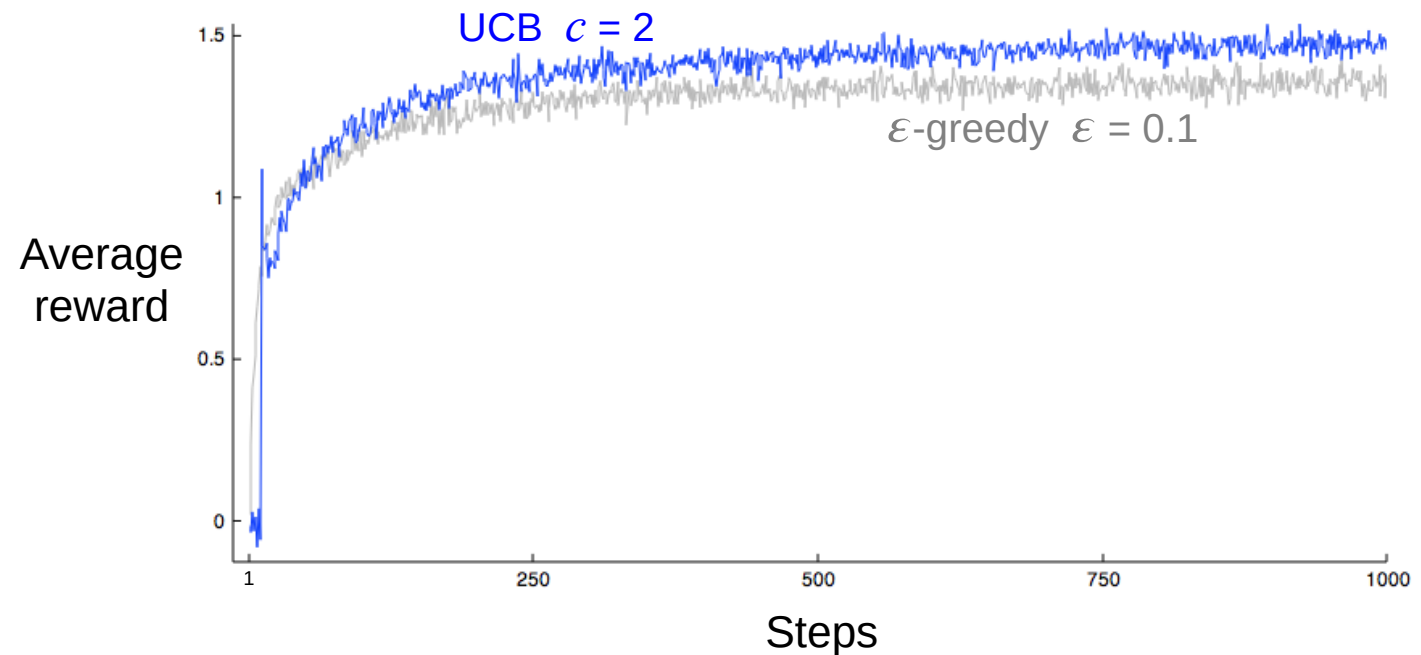
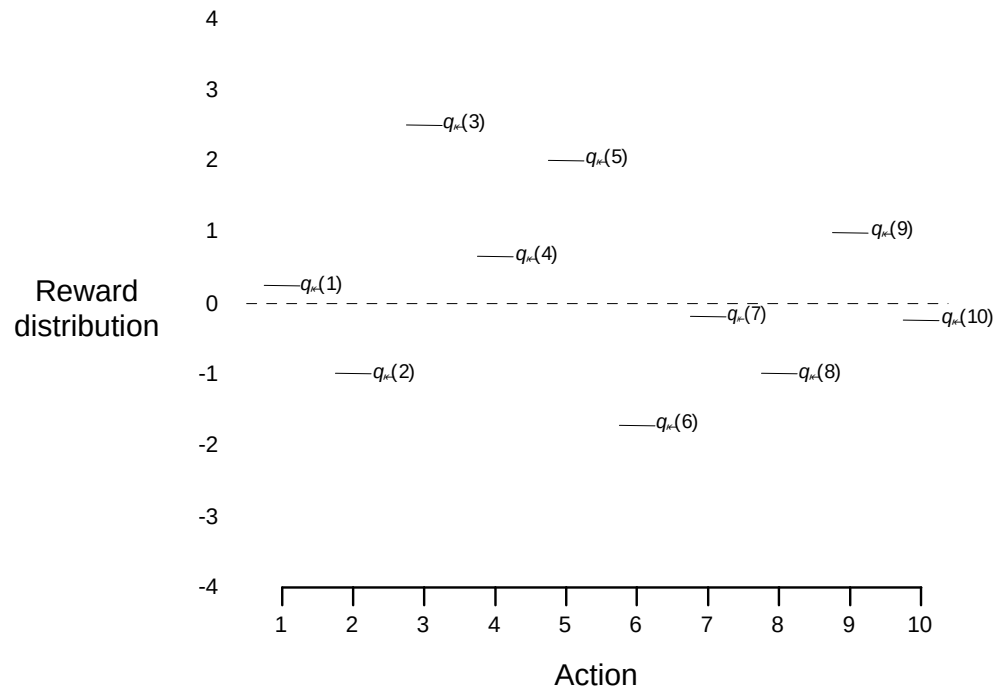
UCB: Upper Confidence Bounds

- ❑ Key idea: choose the most optimistic action, based on $\hat{\mu} + \epsilon$
- ❑ Principles: **optimism in the face of uncertainty** (OFU)
- ❑ UCB will choose action 2 (optimize the optimistic estimates)!



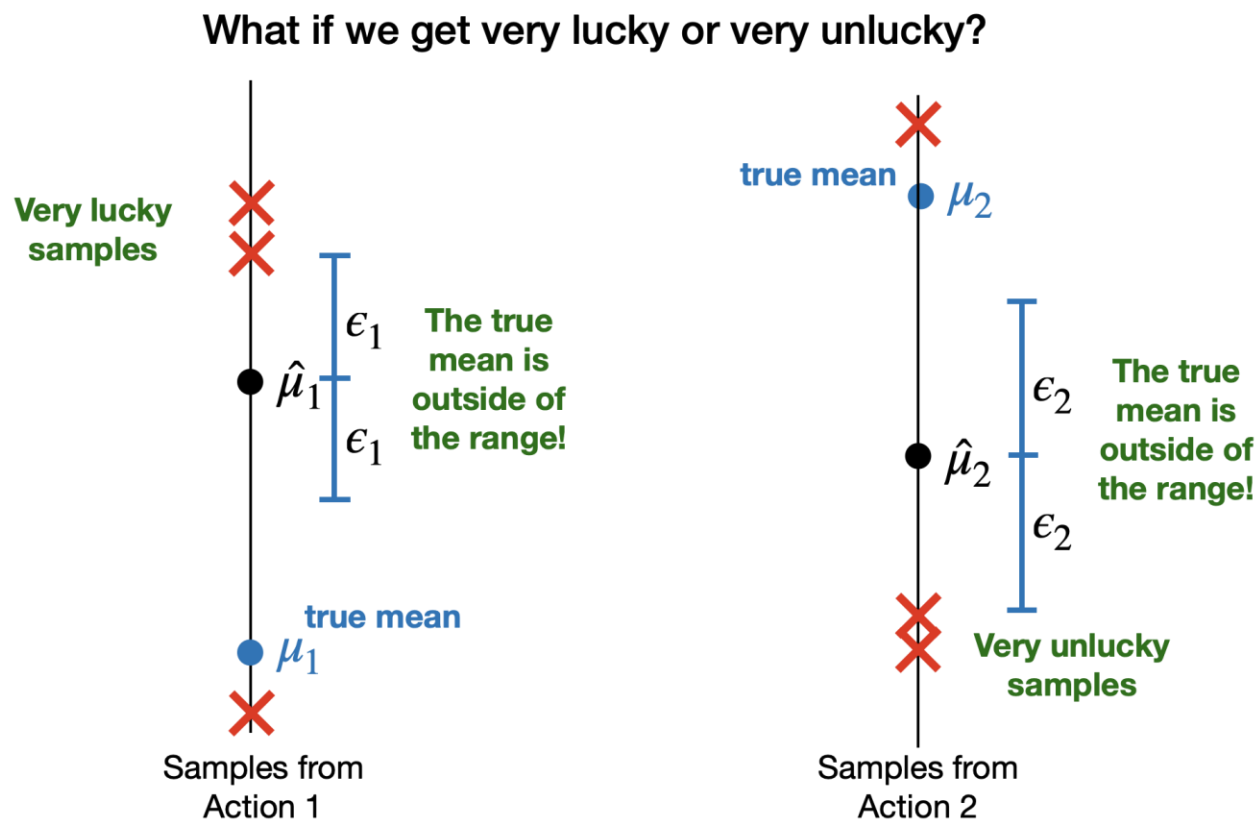
UCB: Upper Confidence Bounds

- ❑ UCB is a theoretically sound algorithm
 - It enjoys sublinear regret! $O(\sqrt{T \cdot \log T})$
- ❑ Works well in practice!



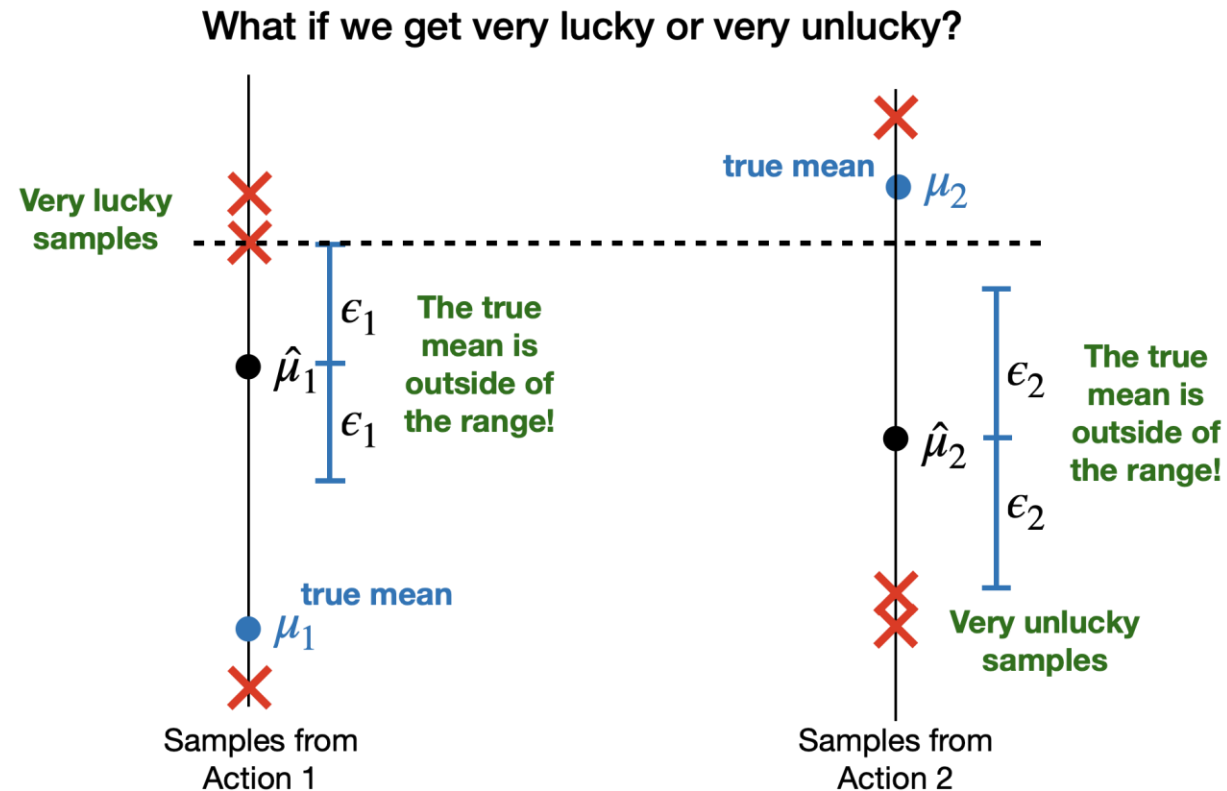
Why UCB Works?

- ❑ Why does UCB NOT get stuck in local optimum?
- ❑ Why can UCB balance exploration and exploitation?



Which action will UCB pick here?

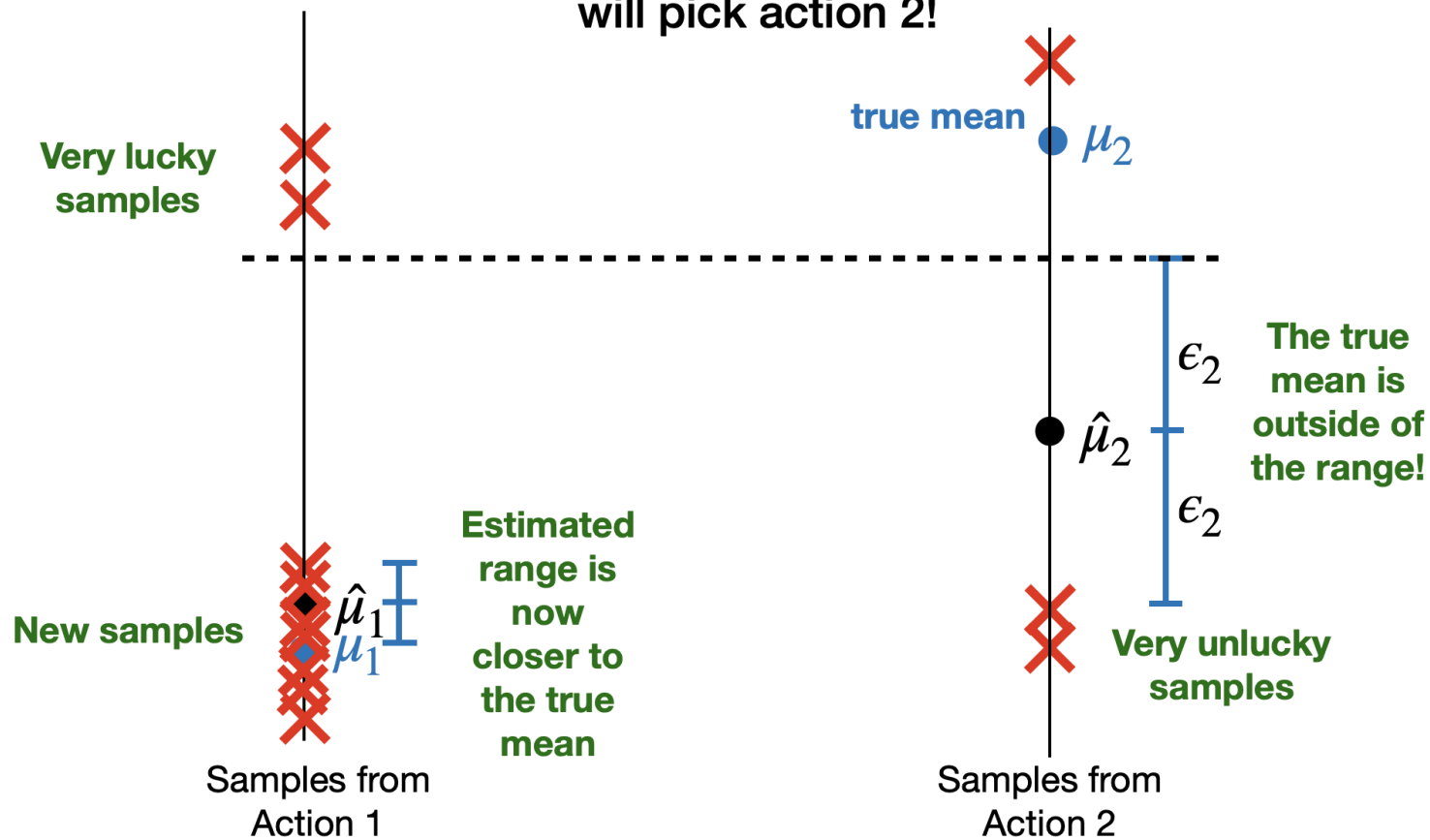
Why UCB Works?



UCB will pick action 1 even though it has a much lower true mean μ_1

- ❑ What will likely happen next? Keep picking the wrong action 1 forever? Or eventually switch to action 2?

The range for action 1 is now corrected and UCB will pick action 2!



The optimistic estimate of the mean is self-correcting!

Thompson Sampling

- ❑ Also unknown as posterior sampling
- ❑ An old and simple idea (1933), but works extremely well in practice
- ❑ No theoretical understanding until recently (Agrawal & Goyal; COLT 2012)

ON THE LIKELIHOOD THAT ONE UNKNOWN PROBABILITY EXCEEDS ANOTHER IN VIEW OF THE EVIDENCE OF TWO SAMPLES.

By WILLIAM R. THOMPSON. From the Department of Pathology,
Yale University.

Section 1.

IN elaborating the relations of the present communication interest was not centred upon the interpretation of particular data, but grew out of a general interest in problems of research planning. From this point of view there can be no objection to the use of data, however meagre, as a guide to action required before more can be collected; although serious objection can otherwise be raised to argument based upon a small number of observations. Indeed, the fact that such objection can never be eliminated entirely—no matter how great the number of observations—suggested the possible value of seeking other modes of operation than that of taking a large number of observations before analysis or any attempt to direct our course. This problem is more general than that treated in *Section 2*, and is directly concerned with any case where probability criteria may be established by means of which we judge whether one mode of operation is *better* than another in some given sense or not.

Thus, if, in this sense, P is the probability estimate that one *treatment* of a certain class of individuals is *better* than a second, as judged by data at present available, then we might take some monotone increasing function of P , say $f_{(P)}$, to fix the fraction of such individuals to be treated in the *first manner*; until more evidence may be utilised, where $0 \leq f_{(P)} \leq 1$; the remaining fraction of such individuals $(1 - f_{(P)})$ to be treated in the *second manner*; or we may establish a probability of treatment by the two methods of $f_{(P)}$ and $1 - f_{(P)}$, respectively. If such a discipline were adopted, even though it were not the best possible, it seems apparent that a considerable saving of individuals otherwise sacrificed to the inferior treatment might be effected. This would be important in cases where either the rate of accumulation of data is slow or the individuals treated are valuable, or both.

If we arbitrarily decide to eliminate the second treatment in favour of the first at this time, then the expectation of sacrifice to the inferior treatment would be $(1 - P)$ for *all* subsequently treated individuals; whereas, if, for example, we take $f_{(P)} = P$, the expectation of such sacrifice would be *temporarily*

$$P(1 - P) + (1 - P)P = 2PQ,$$

On the Likelihood that One Unknown Probability Exceeds Another in View of the Evidence of Two Samples

Author(s): William R. Thompson

Source: *Biometrika*, Vol. 25, No. 3/4 (Dec., 1933), pp. 285-294

Published by: Oxford University Press on behalf of Biometrika Trust

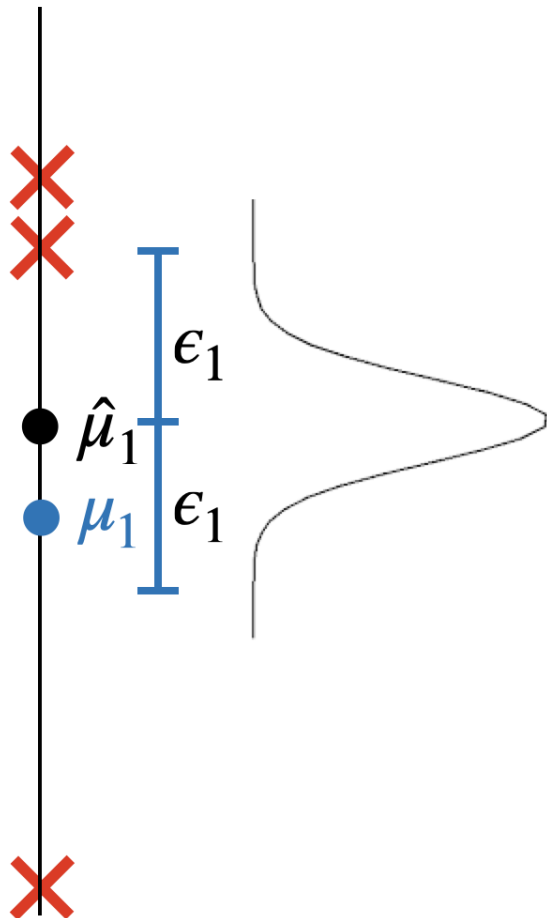
Stable URL: <http://www.jstor.org/stable/2332288>

Accessed: 31-08-2015 03:29 UTC

1933!

Thompson Sampling

- ❑ UCB: estimate a range that the mean lies within, with high probability
- ❑ Thompson sampling: estimate a posterior distribution over μ (NOT a range)



Thompson Sampling does the following:

1) Estimate the posteriors for each arm:

$$p(\mu_i | r_1, \dots, r_n) = \mathcal{N}(\mu'_i, \sigma'_i) \text{ where } r_1, \dots, r_n \text{ are only the rewards for arm } i$$

2) For each arm, sample a mean from its posterior distribution:

$$\hat{\mu}_i \sim \mathcal{N}(\mu'_i, \sigma'_i)$$

3) Choose the arm with the highest **sampled** mean

4) Take the action, observe the reward and go back to step 1

It is why it's also
unknown as
posterior sampling

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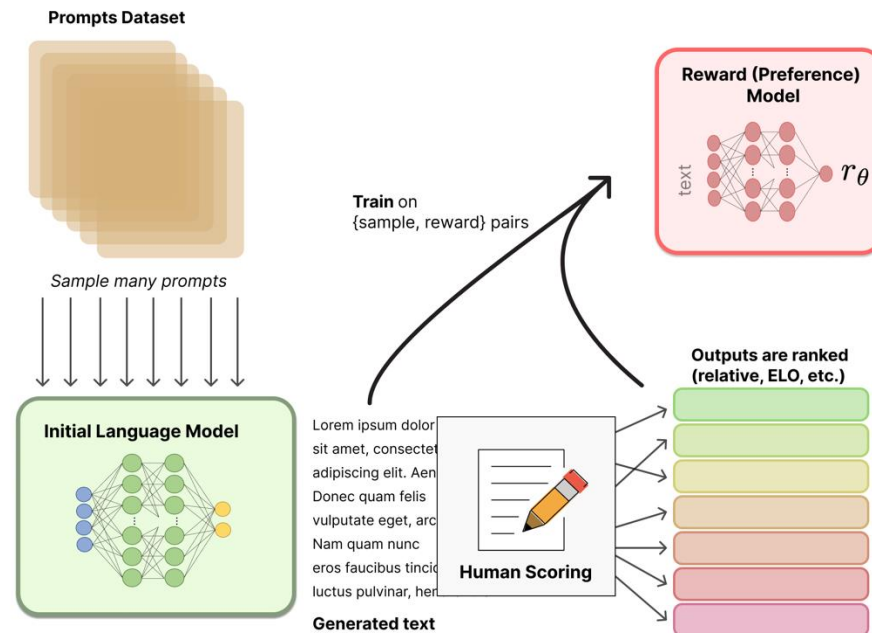
Contextual Bandits

❑ Standard multi-armed bandits:

- $r(a)$: reward of action a
- $Q(a)$: estimation of the mean of $r(a)$

❑ Contextual bandits:

- $r(s, a)$ and $Q(s, a)$ depends on the “context” s
- Similar to state in RL, but no transition/dynamics!



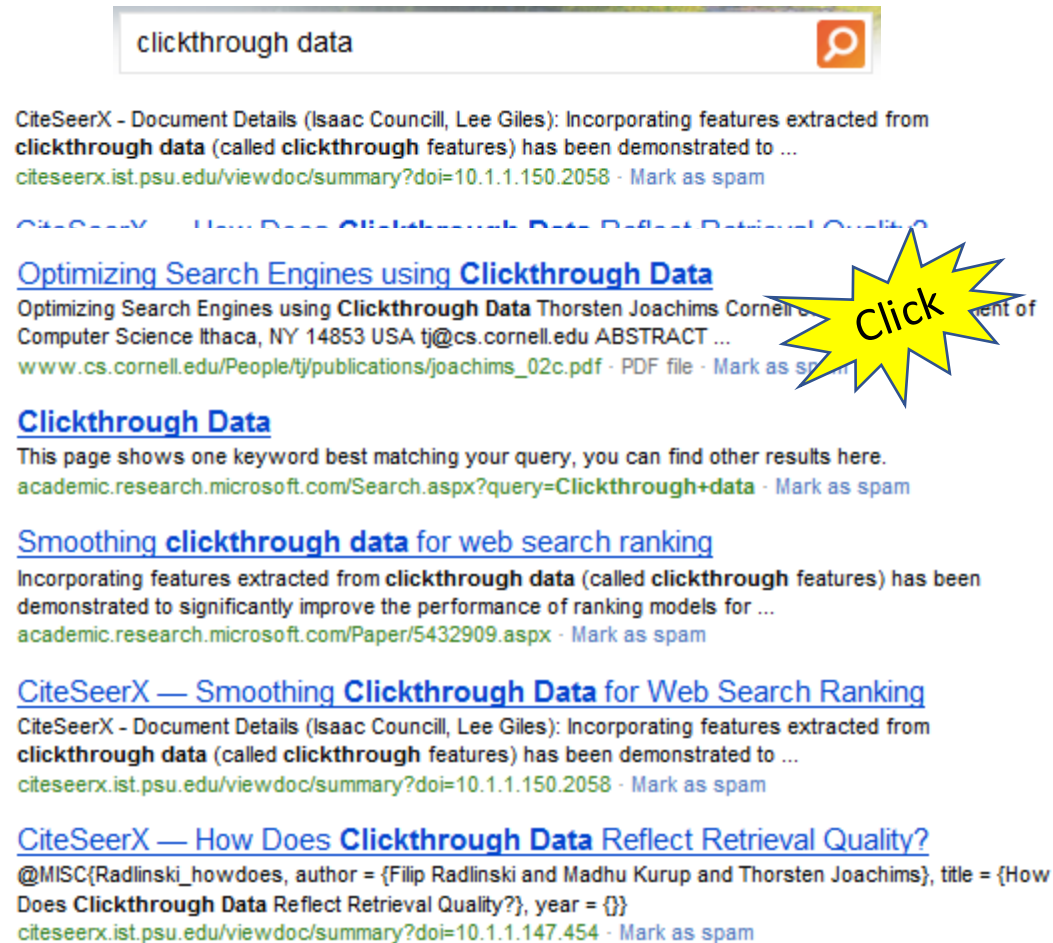
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Preference-based Bandits

- ❑ Motivations: In many applications rewards aren't directly measureable!
- ❑ Or, absolute feedback is not reliable



clickthrough data

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Interpretation 1:

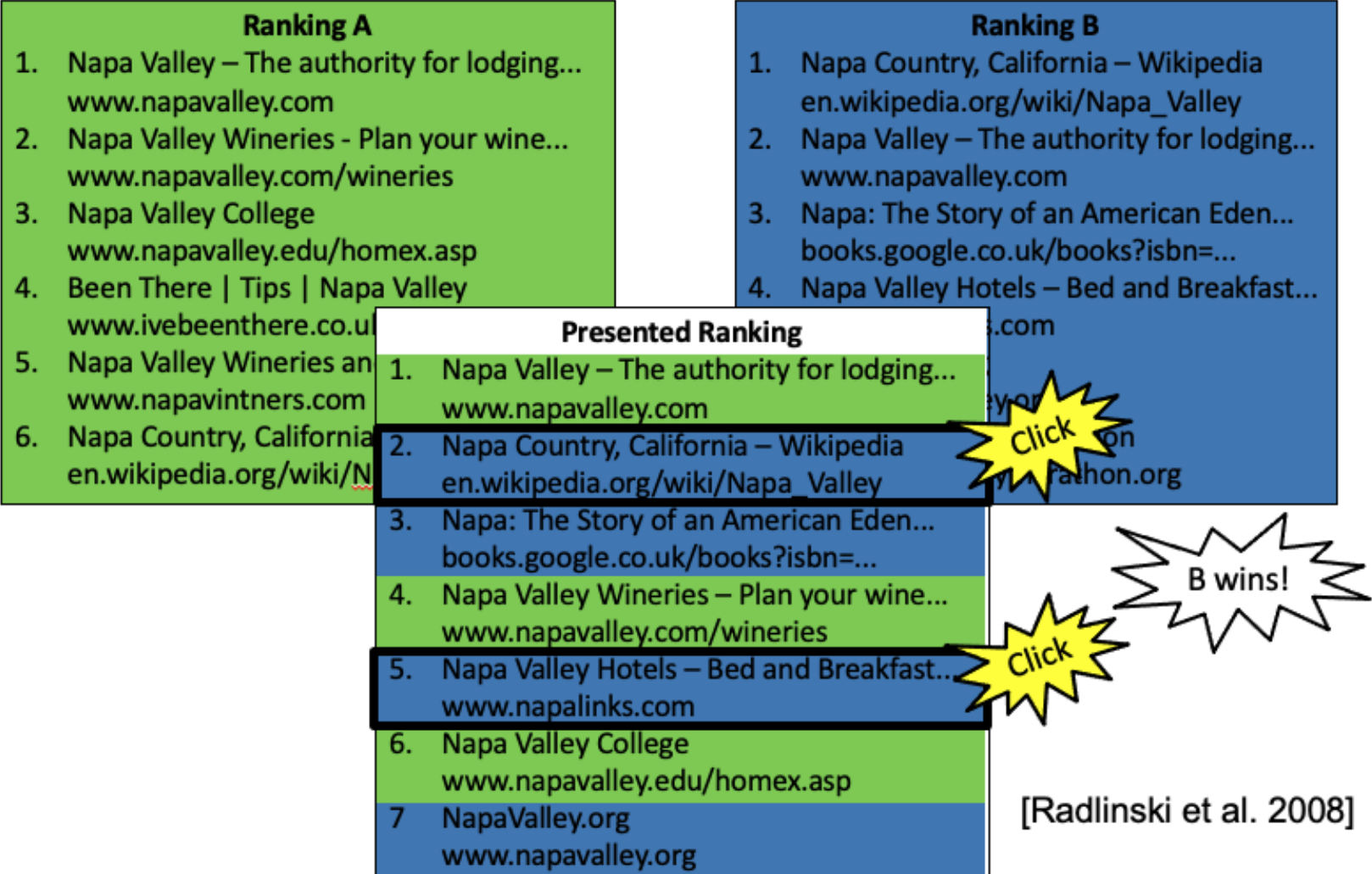
Result #2 is good.
(Absolute)

Interpretation 2:

Result #2 is better
than Result #1.
(Relative / Preference)

Preference-based Bandits

❑ Preference-based feedback is easier to obtain and often more reliable!



Preference-based Bandits

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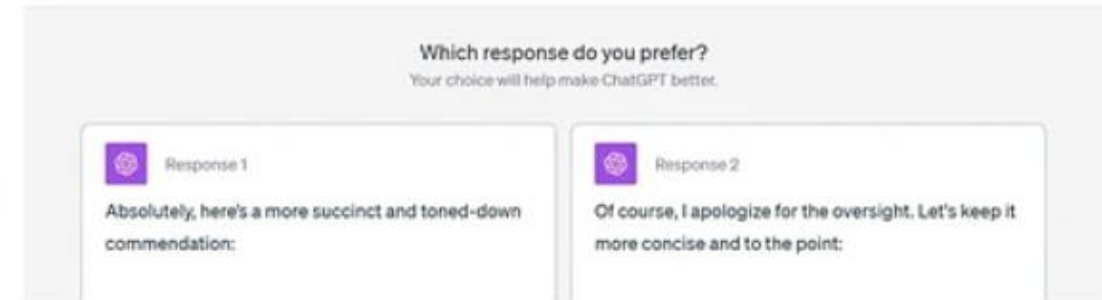
Preference-Based Learning for Exoskeleton Gait Optimization

Maegan Tucker^{*1}, Ellen Novoseller^{*2}, Claudia Kann¹,
Yanan Sui³, Yisong Yue², Joel W. Burdick^{1,2}, and Aaron D. Ames^{1,2}



Preference-based Learning

- ❑ Modern LLMs ask for preference feedback



- ❑ For algorithms, check out the NeurIPS'23 tutorial
 - <https://sites.google.com/view/pref-learning-tutorial-neurips/home>

Do You Prefer Learning with Preferences

Neurips'23 - Dec 11, New Orleans, USA

Aadirupa Saha & Aditya Gopalan



Thank You!